William Dennis

PhD Researcher (PrO-AI CDT), University of Bristol
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Research interests: robust and reliable machine learning; evaluation and benchmarking; applied reinforcement learning; healthcare and wearables.

Experience

Honorary Research Associate in Machine Learning — University of Bristol Oct 2024 – Present
Bristol, UK

- Released and active owner of an open-source fault-injection framework used within the department for the wider research community.
- Actively exchanged knowledge with industry stakeholders on fault-tolerance methods for AI systems.

Research Intern — University of Bristol x National University of Singapore Jun 2026 – Jul 2026
Bristol

- Researched reinforcement learning methods for health behaviour change, supervised by Dr. Mengyan Zhang (University of Bristol) and Swapnil Mishra (National University of Singapore).
- Sole developer of rl-health-interventions: a config-first simulator for Just-In-Time Adaptive Interventions.

Energy Modeller — Aurora Energy Research Oct 2024 – Jun 2026
Oxford, UK

- Forecasted the impact of AI datacentre demand and collaborated with clients interested in SMRs.
- Applied modelling methods using convex optimisation, linear programming, and maximum likelihood estimation with a core focus on explainability, robustness and interpretability.
- Researched and presented a business case for reinforcement learning to improve energy pricing.
- Led modelling of the Polish, Czech and Slovak energy market models, tailoring foundational models to regional needs and building internal toolkits across Europe with Streamlit and SQL.
- Applied best software engineering practices, including test-driven development, Git version control, and agile/scrum workflows.

Data Scientist — JPMorgan Feb 2024 – Jul 2024
Bristol, UK

- Led a team of 4 data scientists to produce in-house quant level research into the development of trading signals derived from millisecond level limit-order-book data.
- Built tooling for my team to parse millions of rows of raw text data on local machines using JIT.
- Created baseline models using ARIMA, RNNs and LSTMs on level-1 data.
- Engineered 50+ features derived from cleaned up level-2 limit order book data.

Signal Processing Engineer — Milbotix Feb 2024 – Oct 2024
Bristol Robotics Lab, UK

- Developed a two-stage machine learning pipeline optimised for use on embedded devices using accelerometer filtering and novel Fourier features.
- Reduced pipeline latency from minutes to seconds via an accelerometer-based filtering layer and JIT compilation.
- Achieved an error rate of <3bpm, allowing the device to be medical grade.
- Extended to lead validation for medical-grade labelling.

Intern Hedging Analyst (Risk Team) — Ocean Partners Jul 2021 – Jul 2022
Maidenhead, UK

- Mitigated £250m of physical commodity transactions via the London Metal Exchange.
- Maintained a strong relationship with banks and brokers using responsive and professional communication.
- Automated a data analysis task for clearing transactions, saving 60 minutes of manual labour daily.

Education

University of Bristol — PhD, PrO-AI CDT Sep 2026 – Present

Doctoral Researcher

- Funded doctoral place under Professor Peter Flach, Department of Computer Science.

University of Bristol — MSc Data Science Sep 2023 – Sep 2024

Distinction (Top 3 of cohort)

- Researched theory of neural network generalisation: grokking, double descent, Lipschitz continuity.
- Deployed distributed cloud architectures on AWS (DynamoDB, SQS, SNS, S3, EC2).
- Studied technology ethics, legal awareness (GDPR), NLP, and the history of technological innovation.
- Produced a Tableau report analysing migration patterns of UK residents aged 20-25.
- Active member of Google Developer Student Club; Quantitative Analyst, Bristol Trading Society.

University of Bath — BSc (Hons) Mathematical Sciences Sep 2019 – Jul 2023

Scientific Computing (84%) and Machine Learning (91%)

- Built websites in Python to support startups in SETsquared using Django.
- 2019/2020 poker society winner.

Research

Robust ML in Harsh Environments 2025

A

novel framework in Python with PyTorch for evaluating neural networks in harsh environments. Proposed two novel activation functions that proved more robust to SEE-related errors.

Framework (GitHub) | Interactive Explanation

Reinforcement Learning for Health Behaviour Change 2026

C

onfig-first simulation environment for Just-in-Time Adaptive Interventions (JITAIs) with bandit and RL baselines, heterogeneous user models, and an LLM user simulator.

GitHub

SLM Bias Testing 2026

B

enchmarking framework for bias in small language models, covering StereoSet, WinoBias, CV screening, and demographic output-length disparity. Evaluates how bias changes across model size, family, and release date.

GitHub

Classification of Signal Quality from Ankle-Based PPG Signals 2024–2025

R

esearch project with Milbotix developing novel time-series features and neural-network classifiers for ankle-based PPG signal-quality classification, evaluated with AUC and ROC metrics.

Publication

Publications

1. **William Dennis** and James Pope. *Robust Machine Learning for Harsh Environments: A Framework and Evaluation of Key Architectural Choices.* Lecture Notes in Computer Science, Vol. 16517 (LNAI), Jun 2026. Published. https://doi.org/10.1007/978-3-032-25032-2_10
2. **William Dennis** and James Pope. *A Framework for Developing Robust Machine Learning Models in Harsh Environments: A Review of CNN Design Choices.* ICAART 2025 (17th International Conference on Agents and Artificial Intelligence), Feb 2025. Published. https://research-information.bris.ac.uk/en/publications/a-framework-for-developing-robust-machine-learning-models-in-hars/
3. Steer, Z., Venkatesh, P. T., Mejia-Mejia, E., **Dennis, W. W.**, Ogundele, P. A., Brooking, A., & Eimontaite, I. *SmartSocks: a new data collection paradigm for dementia and other neurological disorders.* Alzheimer's & Dementia, 20(S8), e095065, 2025. Published. https://doi.org/10.1002/alz.095065

Technical Skills

- **Languages:** Python (10+ years), SQL, C#, R, MATLAB, Haskell, TypeScript, GAMS

- **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, Streamlit, Django
- **Cloud & Tools:** AWS, Google Cloud, Docker, Git, JIRA, Agile

Selected Technical Projects

Moonboard-Analysis 2025-2026

M

L benchmarking framework for climbing route prediction. Designed a reproducible evaluation protocol with 5-fold retrain-per-fold cross-validation, tested 8+ model architectures (MLP, LSTM, CNN, Transformer, Ridge, Perceptron), and established a non-tree leaderboard. Built dataset augmentation pipeline with hold-swap tokenisation and identified/patched data leakage bugs.

GitHub | Benchmark Results

OpenFrontBench 2026

O

pen-source RTS agent benchmark built on a pinned OpenFrontIO engine, with scripted and MCP-based agents, replay verification, and analytical policy baselines.

GitHub

RL Trader 2023

R

einforcement-learning trading agent using linear features.

Activities

- Presented at ICAART 2025 — 17th International Conference on Agents and Artificial Intelligence, Feb 2025.
- Attended Pre-ICML London 2025 — UCL Bloomsbury Campus, Jul 2025.
- Attended AE Global Summit on Open Problems for AI — Friends House, London, Oct 2025.
- Attended UK Theoretical Neuroscience Workshop, Dec 2025.